

RFC-0003

Search Map Specification

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Status: Experimental

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Abstract

This document defines the Search Map format.

A Search Map is the canonical graph representation of an Investigation.

The purpose of a Search Map is not visualization alone.

Its purpose is to expose:

- relationships
- dependencies
- contradictions
- narratives
- source lineages
- knowledge gaps

A Search Map transforms information into navigable structure.

1. Design Philosophy

Traditional systems produce documents.

Searchlores produces maps.

A document is linear.

Knowledge is not.

Knowledge forms networks.

Therefore the canonical representation of an Investigation SHALL be a graph.

2. Core Principle

The fundamental object of a Search Map is:

Node

The fundamental relationship is:

Edge

Everything else derives from these two primitives.

3. Search Map Model

A Search Map consists of:

```
search_map:
  nodes:
  edges:
```

4. Node Definition

A Node represents a knowledge object.

Every node MUST have:

```
node:
  id:
  type:
```

label:

Optional:

description:

metadata:

confidence:

5. Node Types

Official node types:

subject

vector

assumption

contradiction

myth

source

question

narrative

concept

actor

evidence

artifact

Implementations MAY define custom types.

6. Subject Nodes

Purpose:

Represent the root investigation target.

Example:

```
id: llm
type: subject
label: Large Language Models
```

7. Vector Nodes

Purpose:

Represent investigative directions.

Example:

```
id: cognition
type: vector
label: Cognition
```

8. Assumption Nodes

Purpose:

Represent hidden premises.

Example:

```
id: assumption_understanding
type: assumption
label: AI understands language
```

9. Contradiction Nodes

Purpose:

Represent unresolved conflicts.

Example:

```
id: contradiction_reasoning  
type: contradiction  
label: Understanding vs Prediction
```

10. Source Nodes

Purpose:

Represent information origins.

Example:

```
id: source_arxiv  
type: source  
label: arXiv Paper
```

11. Narrative Nodes

Purpose:

Represent explanatory stories.

Example:

```
id: narrative_utopian  
type: narrative  
label: AI Will Save Humanity
```

12. Question Nodes

Purpose:

Represent unresolved inquiry.

Example:

```
id: question_benefits
type: question
label: Who Benefits?
```

Questions are first-class entities.

They are not annotations.

They are map objects.

13. Edge Definition

An Edge defines a relationship.

Every edge MUST contain:

```
edge:
  source:
  target:
  relationship:
```

Optional:

```
weight:
notes:
```

14. Relationship Types

Official relationships:

```
contains  
supports  
contradicts  
questions  
expands_to  
originates_from  
depends_on  
references  
influences  
derived_from
```

15. Supports

Meaning:

Evidence supports a proposition.

Example:

```
source: evidence_1  
target: claim_1  
relationship: supports
```

16. Contradicts

Meaning:

Two nodes are incompatible.

Example:

```
source: claim_a  
target: claim_b  
relationship: contradicts
```

17. Originates From

Meaning:

Genealogy relationship.

Example:

```
source: idea  
target: source  
relationship: originates_from
```

Purpose:

Knowledge lineage reconstruction.

18. Expands To

Meaning:

Investigation branching.

Example:

```
source: llm  
target: economics  
relationship: expands_to
```

19. Search Tree

The simplest Search Map.



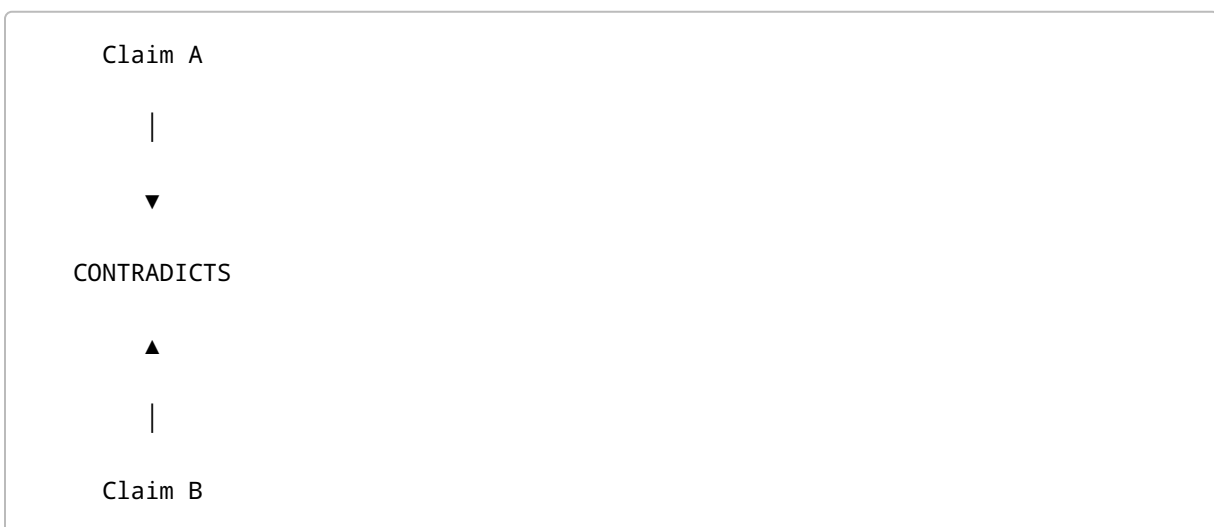
20. Contradiction Graph

Specialized map.

Purpose:

Expose unresolved tensions.

Example:



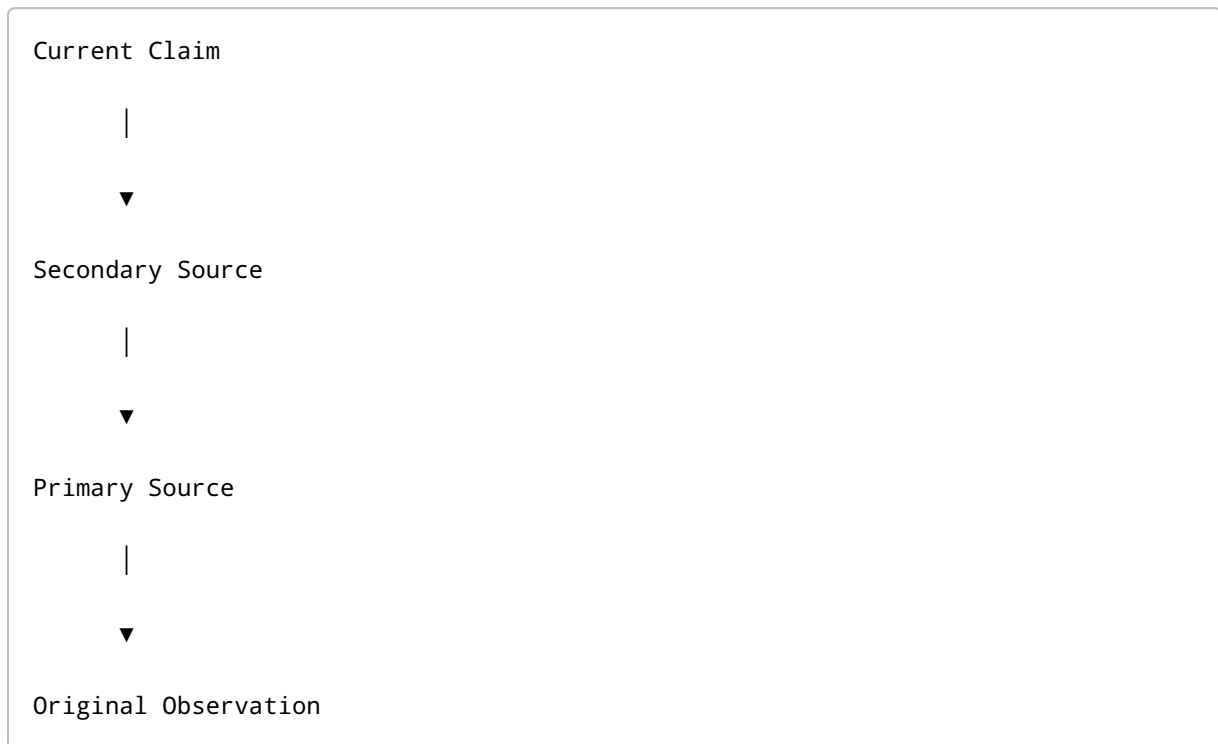
21. Source Tree

Specialized map.

Purpose:

Trace information genealogy.

Example:

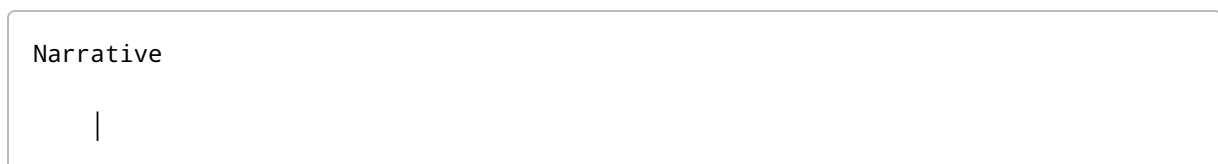


22. Narrative Graph

Purpose:

Map stories.

Example:



└ Promises

|

└ Actors

|

└ Assumptions

|

└ Omissions

23. Question Network

Purpose:

Represent unknowns.

Questions MAY connect to:

- assumptions
- sources
- contradictions
- narratives

Example:

Question

|

└ Source Gap

|

└ Missing Evidence

|

└─ Contradiction

24. Cognitive Topology

Search Maps define knowledge topology.

Regions MAY emerge:

Dense Region

Sparse Region

Contradiction Cluster

Narrative Cluster

Source Cluster

These regions indicate investigation priorities.

25. Unknown Nodes

Searchlores explicitly supports ignorance.

Example:

```
id: unknown_origin
type: unknown
label: Unknown Source
```

Unknowns are legitimate objects.

Absence of knowledge is knowledge.

26. Confidence

Nodes MAY contain confidence values.

Example:

```
confidence: 0.72
```

Confidence measures certainty.

Confidence does NOT measure truth.

27. Serialization

Canonical representation:

```
search_map:  
  
  nodes:  
  
  edges:
```

Alternative representations:

```
GraphML  
  
JSON  
  
Mermaid  
  
DOT  
  
GEXF
```

28. Visualization Targets

A Search Map SHOULD support export to:

- SVG
 - HTML
 - Mermaid
 - Graphviz
 - Gephi
 - Cytoscape
-

29. Example Complete Search Map

```
search_map:

nodes:

  - id: llm
    type: subject
    label: Large Language Models

  - id: cognition
    type: vector
    label: Cognition

  - id: economics
    type: vector
    label: Economics

edges:

  - source: llm
    target: cognition
    relationship: expands_to

  - source: llm
    target: economics
    relationship: expands_to
```

30. Philosophy

Documents describe.

Maps reveal.

Reports summarize.

Maps expose structure.

The objective of Searchlores is not to generate explanations.

The objective of Searchlores is to generate cognitive cartography.

Every investigation ultimately becomes a map.